

PROJECT REPORT

Wine Classification System using Machine Learning

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Course: Data Science / Machine Learning

Abstract

This project focuses on building a machine learning model to classify different types of wine based on their chemical properties. The dataset used contains multiple features such as alcohol content, acidity, and color intensity. Various machine learning algorithms including Logistic Regression and Random Forest were applied to predict the wine category. Hyperparameter tuning techniques were used to improve model performance. The final model achieved high accuracy and was integrated into a prediction system. This project demonstrates the complete machine learning pipeline from data preprocessing to deployment-ready prediction.

Introduction

Machine Learning is a subset of Artificial Intelligence that enables systems to learn from data and make predictions without explicit programming. It plays a crucial role in various industries such as healthcare, finance, and automation.

In this project, we aim to develop a classification model that predicts the type of wine based on input features. This helps in understanding how machine learning models can be applied to real-world datasets for predictive analysis.

Problem Statement

The objective of this project is to build a machine learning model that can accurately classify wine into different categories based on its chemical properties.

Dataset Description

The dataset used in this project is the Wine dataset from Scikit-learn. It contains:

- Multiple input features such as:
 - Alcohol
 - Malic acid
 - Ash
 - Color intensity
 - Flavanoids
- Target variable:
 - Wine class (3 categories)

The dataset is clean, structured, and suitable for classification tasks.

Methodology

The following steps were followed to complete the project:

1. Data Loading
2. Data Preprocessing
3. Train-Test Splitting
4. Feature Scaling
5. Model Training
6. Model Evaluation
7. Hyperparameter Tuning
8. Prediction System Development

Models Used

1. Logistic Regression

- A supervised learning algorithm used for classification
- Works well for linearly separable data

2. Random Forest

- Ensemble learning method
- Combines multiple decision trees
- Reduces overfitting and improves accuracy

Model Evaluation Metrics

The models were evaluated using the following metrics:

- **Accuracy:** Overall correctness of predictions
- **Precision:** Correct positive predictions
- **Recall:** Ability to detect all positives
- **F1 Score:** Balance between precision and recall

Results

Model	Accuracy
Logistic Regression	91%
Random Forest	96%

- Random Forest performed better than Logistic Regression
- Hyperparameter tuning further improved performance

Confusion Matrix

The confusion matrix was used to evaluate model performance by comparing actual vs predicted values.

- Shows correct and incorrect classifications
- Helps identify model weaknesses

• Logistic Regression Report:

	precision	recall	f1-score	support
0	0.95	0.95	0.95	19
1	0.86	0.90	0.88	21
2	0.92	0.86	0.89	14
accuracy			0.91	54
macro avg	0.91	0.90	0.91	54
weighted avg	0.91	0.91	0.91	54

• Random Forest Report:

	precision	recall	f1-score	support
0	1.00	0.95	0.97	19
1	0.91	1.00	0.95	21
2	1.00	0.93	0.96	14
accuracy			0.96	54
macro avg	0.97	0.96	0.96	54
weighted avg	0.97	0.96	0.96	54

Hyperparameter Tuning

GridSearchCV was used to optimize the Random Forest model by tuning parameters such as:

- Number of trees (n_estimators)
- Maximum depth (max_depth)

This improved the model's accuracy and generalization.

Prediction System

A prediction function was created where:

- User inputs feature values
- Model predicts the wine class

This converts the model into a **usable system**, not just theoretical code.

Conclusion

- Machine learning models were successfully trained and evaluated
- Random Forest achieved the best performance
- Hyperparameter tuning improved model accuracy
- A working prediction system was developed

This project demonstrates a complete machine learning workflow.

Future Scope

- Use larger real-world datasets
- Apply advanced models (XGBoost, Neural Networks)
- Deploy the model as a web application
- Improve feature engineering

References

- Scikit-learn Documentation
- Machine Learning textbooks and online resources